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In [1]: import streamlit as st
import seaborn as sns
import matplotlib.pyplot as plt

# Set Seaborn style
sns.set(style="whitegrid")

# Load the tips dataset
tips = sns.load_dataset('tips')

# Streamlit app
st.title("Comprehensive Tips Dataset Visualizations")
st.write("This app displays a variety of Seaborn plots created by prakash senapa")

# Function to create and display a plot
def display_plot(title, plot_func):
    st.subheader(title)
    fig, ax = plt.subplots(figsize=(8, 6))
    plot_func(ax)
    st.pyplot(fig)
    plt.close(fig) # Close the figure to free memory

# Plot definitions
def scatter_plot(ax):
    sns.scatterplot(data=tips, x="total_bill", y="tip", hue="time", size="size",
ax.set_title("Scatter Plot: Total Bill vs Tip by Time and Size")

def line_plot(ax):
    sns.lineplot(data=tips, x="size", y="total_bill", hue="sex", ci=None, marker
ax.set_title("Line Plot: Total Bill by Party Size and Sex")

def bar_plot(ax):
    sns.barplot(data=tips, x="day", y="total_bill", hue="sex", palette="muted",
ax.set_title("Bar Plot: Average Total Bill by Day and Sex")

def box_plot(ax):
    sns.boxplot(data=tips, x="day", y="tip", hue="smoker", palette="Set2", ax=ax
ax.set_title("Box Plot: Tips by Day and Smoker Status")

def violin_plot(ax):
    sns.violinplot(data=tips, x="day", y="total_bill", hue="time", split=True, p
ax.set_title("Violin Plot: Total Bill by Day and Time")

def count_plot(ax):
    sns.countplot(data=tips, x="day", hue="smoker", palette="dark", ax=ax)
ax.set_title("Count Plot: Orders by Day and Smoker Status")

def reg_plot(ax):
    sns.regplot(data=tips, x="total_bill", y="tip", scatter_kws={"s": 50}, line_
ax.set_title("Regression Plot: Total Bill vs Tip")

def hist_plot(ax):
    sns.histplot(data=tips, x="total_bill", kde=True, bins=20, color="blue", ax=
ax.set_title("Histogram: Distribution of Total Bill")

def strip_plot(ax):
    sns.stripplot(data=tips, x="day", y="tip", hue="sex", palette="Set1", jitter
ax.set_title("Strip Plot: Tips by Day and Sex")
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def swarm_plot(ax):
    sns.swarmplot(data=tips, x="day", y="tip", hue="smoker", palette="Set3", ax=ax)
    ax.set_title("Swarm Plot: Tips by Day and Smoker Status")

def kde_plot(ax):
    sns.kdeplot(data=tips, x="total_bill", hue="sex", fill=True, palette="tab10")
    ax.set_title("KDE Plot: Total Bill Density by Sex")

# Special cases that need custom handling
st.subheader("Pair Plot: Numerical Variables by Sex")
pair_fig = sns.pairplot(tips, hue="sex", vars=["total_bill", "tip", "size"], palette="Set3")
pair_fig.fig.suptitle("Pair Plot: Numerical Variables by Sex", y=1.02)
st.pyplot(pair_fig.fig)
plt.close(pair_fig.fig)

st.subheader("Joint Plot: Total Bill vs Tip by Smoker")
joint_fig = sns.jointplot(data=tips, x="total_bill", y="tip", kind="scatter", hue="smoker")
joint_fig.fig.suptitle("Joint Plot: Total Bill vs Tip by Smoker", y=1.02)
st.pyplot(joint_fig.fig)
plt.close(joint_fig.fig)

st.subheader("FacetGrid: Total Bill by Time and Smoker")
g = sns.FacetGrid(tips, col="time", row="smoker", margin_titles=True)
g.map(sns.histplot, "total_bill", bins=15)
g.fig.suptitle("FacetGrid: Total Bill by Time and Smoker", y=1.02)
st.pyplot(g.fig)
plt.close(g.fig)

st.subheader("Catplot (Point): Tips by Day and Sex")
cat_fig = sns.catplot(data=tips, x="day", y="tip", hue="sex", kind="point", palette="Set3")
cat_fig.fig.suptitle("Catplot (Point): Tips by Day and Sex", y=1.02)
st.pyplot(cat_fig.fig)
plt.close(cat_fig.fig)

# Display all standard plots
display_plot("Scatter Plot", scatter_plot)
display_plot("Line Plot", line_plot)
display_plot("Bar Plot", bar_plot)
display_plot("Box Plot", box_plot)
display_plot("Violin Plot", violin_plot)
display_plot("Count Plot", count_plot)
display_plot("Regression Plot", reg_plot)
display_plot("Histogram", hist_plot)
display_plot("Strip Plot", strip_plot)
display_plot("Swarm Plot", swarm_plot)
display_plot("KDE Plot", kde_plot)

st.write("All Seaborn plots for the Tips dataset have been generated!")

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2025-09-06 19:09:27.610 WARNING streamlit.runtime.scriptrunner_utils.script_run_c
ontext: Thread 'MainThread': missing ScriptRunContext! This warning can be ignore
d when running in bare mode.
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2025-09-06 19:09:28.034
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Warning: to view this Streamlit app on a browser, run it with the following command:

```
streamlit run C:\Users\IPL4\anaconda3\lib\site-packages\ipykernel_launcher.py
[ARGUMENTS]
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C:\Users\IPL4\AppData\Local\Temp\ipykernel_13520\4168873102.py:29: FutureWarning:

```

The `ci` parameter is deprecated. Use `errorbar=None` for the same effect.

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sns.lineplot(data=tips, x="size", y="total_bill", hue="sex", ci=None, marker
="o", ax=ax)

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In []: